



KARAN RAJ T

AI/ML Engineer | LLM & RAG Systems

PROFESSIONAL SUMMARY

AI and Data Science undergraduate specializing in Large Language Models, Retrieval-Augmented Generation, and applied machine learning. Builds local-first AI agents integrating multiple LLM providers, with hands-on experience in RAG pipelines, semantic search, agentic automation, and classical AI search algorithms.

EDUCATION

B.Tech, Artificial Intelligence and Data Science — Mepco Schlenk Engineering College, Sivakasi Pre-Final Year (2024–2028) | CGPA: 8.39/10
Higher Secondary Certificate (Class XII) — S.H.N. Edward Higher Secondary School, Sattur | 87.67% (2024)
Secondary School Leaving Certificate (Class X) — S.H.N. Edward Higher Secondary School, Sattur | 85.80% (2022)

PROJECTS

VIP Assistant — Local AI Agent with Advanced RAG *Jul 2026 – Present*
Python, Node.js, React, Ollama, Hugging Face, RAG, Vector Database, WebSockets

- Built a local AI assistant integrating cloud and on-device LLMs with multi-provider support (Ollama, Gemini, Claude, NVIDIA NIM).
- Implemented Retrieval-Augmented Generation with semantic search and workspace indexing for intelligent code assistance.

Smart Beneficiary Mapping System *Feb 2026 – Present*
Python, Flask, MySQL, AI Agents, Automation, Rule Engine

- Designed an AI-enhanced eligibility recommendation platform for government welfare schemes, evolving from a rule-based engine to an autonomous agent-driven system.
- Added autonomous agent workflows and intelligent recommendations to reduce manual effort and improve accessibility.

Stock Market Analysis & Prediction *Apr 2026 – May 2026*
Python, Pandas, Scikit-learn, Streamlit, Machine Learning

- Built an interactive dashboard for stock market analysis and prediction using multiple ML algorithms, including regression, classification, and model comparison.

Maze Runner 3D using AI Search Techniques *Mar 2026 – May 2026*
Python, Flask, JavaScript, Three.js, Artificial Intelligence

- Developed a 3D maze game demonstrating classical AI search algorithms with intelligent pathfinding and obstacle navigation.

LEADERSHIP & RESPONSIBILITIES

Google Student Club — Office Bearer *2025 – Present*
The Institution of Engineers (India) — Executive Member *2025 – Present*
AIMEE Association — Design and Publicity Member *2026 – Present*
Scout — Scout Leader *2020 – 2022*

MERIT & RECOGNITION

- Certificate of Merit — Executive Member, The Institution of Engineers (India) – AI & DS Student Chapter (2025–2026).
- Certificate of Merit — Office Bearer, Google Student Club (2025–2026).
- Presented Smart Beneficiary Mapping Platform at TECHNOINNOVATE 2025–2026, IIC, Mepco Schlenk Engineering College.

CONTACT

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TECHNICAL SKILLS

AI/ML
NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Hugging Face, Ollama, RAG, Vector Databases, LLM Fine-Tuning
Languages
Python, C, C++, Core Java
Databases & Tools
MySQL, Oracle SQL, Git, GitHub, VS Code, Jupyter Notebook, Power BI

AREAS OF INTEREST

Large Language Models, Hugging Face, Llama.cpp, Retrieval-Augmented Generation, LLM Fine-Tuning, AI Model Optimization

CERTIFICATIONS

- Introduction to Internet of Things (Elite) — NPTEL, IIT Kharagpur | Jan–Apr 2026
- Design & Implementation of Human-Computer Interfaces — NPTEL, IIT Guwahati | Jul–Oct 2025
- IEEE English for Technical Professionals — IEEE | May 2025

ADDITIONAL INFORMATION

Coding Profiles
LeetCode, HackerRank, CodeChef
Languages
English, Tamil